$$r = \frac{i}{F} = C_R k_a \exp\left[\frac{(1-\beta)FV}{RT}\right] - C_o k_c \exp\left[\frac{-\beta FV}{RT}\right]$$

We characterize exchange current density as rate of forward (or reverse) rxn when overall rxn is @ equil
↳ We define the potential @ which this occurs as U

$$i \Rightarrow 0 \Rightarrow \quad \cancel{F} C_R k_a \exp\left[\frac{(1-\beta)FU}{RT}\right] = \cancel{F} C_o k_c \exp\left[\frac{-\beta FU}{RT}\right] = i_o$$

↓

Solving for U:

$$\exp\left[\left(\frac{(1-\beta)F}{RT} + \frac{\beta F}{RT}\right)U\right] = \frac{C_o k_c}{C_R k_a}$$

$$U = \frac{\ln\left(\frac{C_o k_c}{C_R k_a}\right)}{\frac{(1-\beta)F}{RT} + \frac{\beta F}{RT}} = \ln\left(\frac{C_o k_c}{C_R k_a}\right)\frac{RT}{F}$$

$$\therefore \boxed{U = \frac{RT}{F}\ln\left(\frac{k_c}{k_a}\right) + \frac{RT}{F}\ln\left(\frac{C_o}{C_R}\right)}$$

Introduce: $V = U + \eta$
↑ Overpotential → plugging into i expression

$$\therefore i = \underbrace{F C_R k_a \exp\left[\frac{(1-\beta)UF}{RT}\right]}_{i_o} \exp\left[\frac{(1-\beta)\eta F}{RT}\right] - \underbrace{F C_o k_c \exp\left[\frac{-\beta FU}{RT}\right]}_{i_o} \exp\left[\frac{-\beta F\eta}{RT}\right]$$

$$\boxed{i = i_o\left(\exp\left[\frac{(1-\beta)F\eta}{RT}\right] - \exp\left[\frac{-\beta F\eta}{RT}\right]\right)} \longrightarrow \text{Butler-Volmer Eq}$$

For rxns where multiple e^- transfers occur, we use α instead of β:

$$i = i_o\left(\exp\left[\frac{\alpha_a F\eta}{RT}\right] - \exp\left[\frac{-\alpha_c F\eta}{RT}\right]\right) \qquad 0 \le \alpha \le n$$

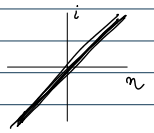
↑ transfer coeff.

Simplifications:

↳ @ small overpotential $\eta \to 0$; linearize B-V w/ 1st order Taylor-expansion

$$\boxed{i = i_o\frac{(\alpha_a + \alpha_c)F\eta}{RT}}$$

linear kinetics:



↳ @ large overpotential, ex $\eta \gg 0$
Total current contribution from cathodic rxn becomes small relative to anodic rxn,

$$i = i_o\left(\exp\left[\frac{\alpha_a F\eta}{RT}\right] - \exp\left[\frac{-\alpha_c F\eta}{RT}\right]\right)$$

positive, large — negative, small

$$\therefore \boxed{i = i_o \exp\left[\frac{\alpha_a F\eta}{RT}\right]}$$ Anodic Tafel equation

large small

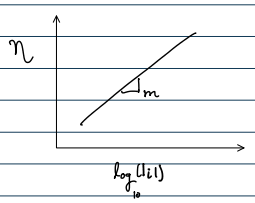
$$\therefore \boxed{i = i_0 \exp\left[\frac{\alpha_a F \eta}{RT}\right]}$$ Anodic Tafel equation

↳ Taking $\log_{i_0}$ to linearize:

$$\log_{10}(i) - \log_{10}(i_0) = \frac{\alpha_a F \eta}{RT} \quad \log_{10}(e) = \frac{\alpha_a F \eta}{2.303 RT}$$

$$\eta = m \cdot \log_{10}(i) + b$$

$m = \frac{2.303 RT}{F \alpha_a}$ $\quad b = -m \log_{10}(|i_0|)$



m = Tafel slope $\left[\frac{mV}{decade}\right]$

* analogous to Arrhenius plot

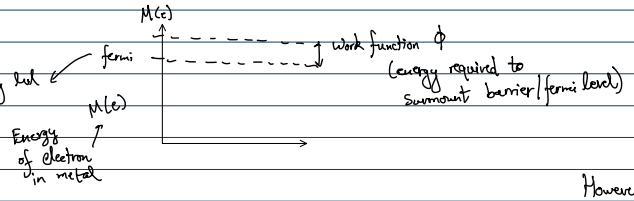
Slope → higher would mean that a small increase in η lead to ↑ i_0 (higher rate?)

Actually, a lower slope means that a small overpotential causes a larger increase in current, therefore smaller slope is higher rate!!! (since overpotential is on the y axis) ✓

→ **β (Symmetry factor)**

Fraction of $F(V_2 - V_1)$ required to overcome energy barrier → What does it physically represent?

+ β characterizes the quantum mechanical transfer of e^- from electrode to chem species
↳ Classical mechanics of e^- predicts orders of mag. lower i than observed for potentials we apply;

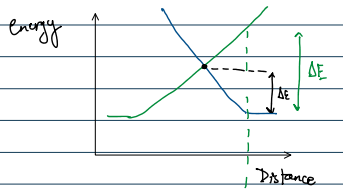


e^- needs to escape metal and then get transferred to another fermi level → However, recorded energies were for lower than predicted

↳ QM allows e^- to tunnel rather than overcome energy barrier → this occurs w/out radiation energy (No light is seen)
↳ this requires initial + final state have to have the same energy

ex: adsorption of proton @ electrode

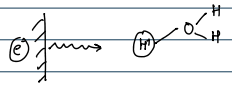
$$M(e^-) + H^+ - OH_2 \rightleftharpoons M-H + H_2O$$



– initial state

– final state

Initial



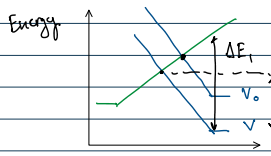
ΔE → large energy barriers req. to transfer e^- from ground state (no tunneling possible)

Δε → represents activation barrier for tunneling

Final:



When we apply a potential we vertically shift initial state & thus change $\Delta\varepsilon$:



$$\Delta(\Delta\varepsilon) = \Delta\varepsilon_1 - \Delta\varepsilon_0$$

$$\frac{\Delta\varepsilon_0}{\Delta E_0} = \frac{\Delta\varepsilon_1}{\Delta E_1} \quad \text{(straight line)}$$

$$= \frac{\Delta\varepsilon_0 + \Delta(\Delta\varepsilon)}{\Delta E_0 + \Delta V F}$$

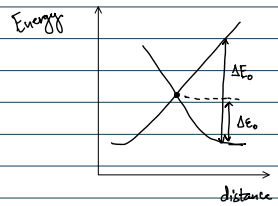
$F(V_1 - V_0)$ gives energy associated w/ voltage change

?

$$\frac{\Delta\varepsilon_0}{\Delta E_0} = \frac{\Delta(\Delta\varepsilon)}{\Delta V F}$$

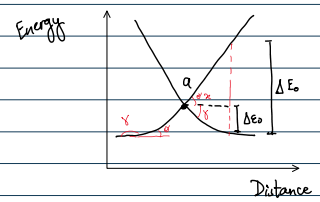
↳ represents the fraction of work applied, that required to change E_a

$$= \beta$$



$$\frac{\Delta\varepsilon_0}{\Delta E_0} = \beta$$

Get value of β from trig.



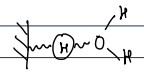
$$\beta = \frac{\Delta\varepsilon_0}{\Delta E_0} = \frac{x \tan\gamma}{x(\tan\gamma + \tan\theta)}$$ } Cathodic

$$(1-\beta) = \frac{\tan\theta}{\tan\gamma + \tan\theta}$$ } Anodic

Straight line assumption → validity?

↳ has to

ⓐ a → excited state



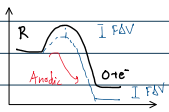
$R \rightleftharpoons O + e^-$

β → Fraction of energy for ΔV change going into activation energy change

If slopes ≈ equal: $\tan\gamma \approx \tan\theta$

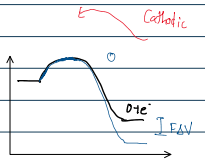
$\beta = \frac{1}{2}$

$\beta \approx 0$



$\beta=0$ → Anodic process is greatly impacted by ΔV

→ Cathodic process is not!



Anodic process is not impacted by ΔV!